Networking Academy

Packet Tracer - VLSM Design and Implementation Practice Topology

You will receive one of three possible topologies.

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Building 1	G0/0	172.31.107.1	255.255.255.224	N/A
	G0/1	172.31.107.33	255.255.255.224	N/A
	S0/0/0	172.31.107.97	255.255.255.252	N/A
Building 2	G0/0	172.31.107.65	255.255.255.240	N/A
	G0/1	172.31.107.81	255.255.255.240	N/A
	S0/0/0	172.31.107.98	255.255.255.252	N/A
ASW-1	VLAN 1	172.31.107.2	255.255.255.224	172.31.107.1
ASW-2	VLAN 1	172.31.107.34	255.255.255.224	172.31.107.33
ASW-3	VLAN 1	172.31.107.66	255.255.255.240	172.31.107.65
ASW-4	VLAN 1	172.31.107.82	255.255.255.240	172.31.107.81
Host-A	NIC	172.31.107.30	255.255.255.224	172.31.107.1
Host-B	NIC	172.31.107.62	255.255.255.224	172.31.107.33
Host-C	NIC	172.31.107.78	255.255.255.240	172.31.107.65
Host-D	NIC	172.31.107.94	255.255.255.240	172.31.107.81

Objectives

Part 1: Examine the Network Requirements

Part 2: Design the VLSM Addressing Scheme

Part 3: Assign IP Addresses to Devices and Verify Connectivity

Background

In this activity, you are given a /24 network address to use to design a VLSM addressing scheme. Based on a set of requirements, you will assign subnets and addressing, configure devices and verify connectivity.

Instructions

Part 1: Examine the Network Requirements

Step 1: Determine the number of subnets needed.

You will subnet the network address 172.31.107.0/24. The network has the following requirements:

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	Subnet Description	Number of Hosts Needed	Network Address/CIDR	First Usable Host Address	Broadcast Address
/27	ASU-1 LAN	27	172.31.107.0	172.31.107.1	172.31.107.31
/27	ASU-2 LAN	25	172.31.107.32	172.31.107.33	172.31.107.63
/25	ASU-3 LAN	14	172.31.107.64	172.31.107.65	172.31.107.79
/28	ASU-4 LAN	6	172.31.107.80	172.31.107.81	172.31.107.95
/30	R1-R2 WAN	2	172.31.107.96	172.31.107.97	172.31.107.99

Step 3: Document the addressing scheme.

a. Assign the first usable IP addresses to Building 1 for the two LAN links and the WAN link.

b. Assign the first usable IP addresses to Building 2 for the two LAN links. Assign the last usable IP address for the WAN link.

c. Assign the second usable IP addresses to the switches.

d. Assign the last usable IP addresses to the hosts.

Part 3: Assign IP Addresses to Devices and Verify Connectivity

Most of the IP addressing is already configured on this network. Implement the following steps to complete the addressing configuration.

Step 1: Configure IP addressing on the Building 1 router LAN interfaces.

Step 2: Configure IP addressing on the ASU-3 , switch including the default gateway.

Step 3: Configure IP addressing on Host-D , including the default gateway.

Step 4: Verify connectivity.

You can only verify connectivity from Building 1 , ASU-3 , and Host-D . However, you should be able to ping every IP address listed in the **Addressing Table**.

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- ASW-1 LAN will require 27 host IP addresses
- ASW-2 LAN will require 25 host IP addresses
- ASW-3 LAN will require 14 host IP addresses
- ASW-4 LAN will require 8 host IP addresses

How many subnets are needed in the network topology? 5

Step 2: Determine the subnet mask information for each subnet.

a. Which subnet mask will accommodate the number of IP addresses required for ASW-1 ?

How many usable host addresses will this subnet support?

/27 30 usable

b. Which subnet mask will accommodate the number of IP addresses required for ASW-2 ?

How many usable host addresses will this subnet support?

/27 30 usable

c. Which subnet mask will accommodate the number of IP addresses required for ASW-3 ?

How many usable host addresses will this subnet support?

/28 14 usable

d. Which subnet mask will accommodate the number of IP addresses required for ASW-4 ?

How many usable host addresses will this subnet support?

/28 14 usable

e. Which subnet mask will accommodate the number of IP addresses required for the connection between Building 1 and Building 2 ? /30

Part 2: Design the VLSM Addressing Scheme

Step 1: Divide the 172.31.107.0/24 network based on the number of hosts per subnet.

a. Use the first subnet to accommodate the largest LAN.

b. Use the second subnet to accommodate the second largest LAN.

c. Use the third subnet to accommodate the third largest LAN.

d. Use the fourth subnet to accommodate the fourth largest LAN.

e. Use the fifth subnet to accommodate the connection between and .

Step 2: Document the VLSM subnets.

Complete the **Subnet Table**, listing the subnet descriptions (e.g. [[S1Name]] LAN), number of hosts needed, then network address for the subnet, the first usable host address, and the broadcast address. Repeat until all addresses are listed.

Subnet Table

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